

Fabulous smoothing: main (MeshLib) vs GeometryEngine

Head-to-head over eleven clinical and synthetic inputs at the UI's standard preset · 27 September 2026

Introduction

Fabulous on main smooths a bolus with MeshLib voxel offsets and a geometry3Sharp Reducer. The geometry-engine branch smooths it with GeometryEngine's own DoubleOffset, Offset and Decimate. The two had never been run on the same input and compared, so this report does that: the same eleven meshes, the same settings, the same frame.

The question is not which is prettier, but whether the engine can replace main without changing the bolus a clinic prints — and, where it does change it, by how much and why.

How this was tested

Two headless harnesses generate the meshes, one per pipeline, each centring its input on the bounding box exactly as its own app does on import. The reference harness carries verbatim copies of main's smoothing code, since Fabulous.Core on main will not build off Windows.

Both run the UI's standard preset: a 1.0 mm offset cycle, 0.1 mm inflation, one iteration, 1.0 mm cell, decimated to twice the input triangle count. Component separation, which only the engine branch does, is off so both sides see the same geometry.

Measured with Fabulous.MeshCompare on linux-x64, against libmanifoldc built at upstream 7c86359 — the commit the shipped win-x64 binaries come from. ICP between the outputs returned under 0.04 mm and 0.18°, so the harnesses agree about where the model is.

Summary

The geometry matches closely: mean Dice 0.992, HD95 between 0.03 and 0.23 mm, mean surface separation under 0.07 mm. The engine sits a touch outside main — bias +0.010 mm on average, never past +0.030 mm — which the code predicts: main's inflation offset runs on a finer grid, and the engine's decimation has no projection target.

Against the unsmoothed anatomy the engine is the closer of the two, halving main's HD95 on nine of eleven cases, while growing volume slightly more.

Topology is the gap: three engine outputs are not watertight against none on main, and a fourth gains two self-intersections. That is what to fix before the engine replaces main. Runtime is 2.1x main's — the offsets cost more, the decimation costs less.

Per-case result

distances in mm; volume and area are the engine's difference from main; W / NM / SI = watertight, non-manifold edges, self-intersections

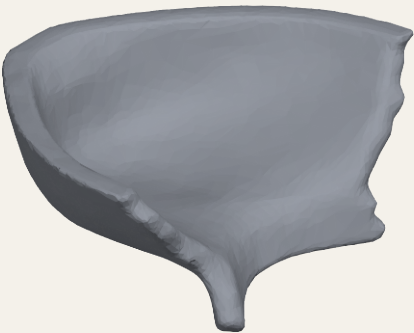
case	in tris	out tris	Dice	HD95	ASSD	bias	vol %	area %	main W / NM / SI	engine W / NM / SI	main s	engine s	checks
chin_bolus	3,216	6,432	0.9907	0.133	0.045	+0.007	+0.31	+1.17	yes / 0 / 0	NO / 3 / 0	1.52	2.92	topology
ear_bolus	2,810	5,624	0.9898	0.183	0.047	+0.012	+0.53	+1.72	yes / 0 / 0	yes / 0 / 0	1.02	2.09	all pass
eye_bolus	1,154	2,308	0.9861	0.226	0.061	+0.029	+1.34	+2.64	yes / 0 / 0	yes / 0 / 0	0.63	0.85	volume, ICP noise
larynx small	2,978	5,960	0.9844	0.218	0.069	+0.016	+0.74	+2.44	yes / 0 / 0	NO / 3 / 0	0.68	2.29	topology
larynx_bolus	10,694	21,388	0.9961	0.121	0.032	+0.002	+0.04	+0.63	yes / 0 / 0	NO / 1 / 0	1.43	2.50	topology
nose_bolus	1,566	3,132	0.9857	0.200	0.067	+0.012	+0.51	+2.34	yes / 0 / 0	yes / 0 / 0	0.67	0.98	all pass
scalp_bolus	8,546	17,092	0.9946	0.086	0.027	+0.004	+0.16	+0.59	yes / 0 / 0	yes / 0 / 0	1.29	3.35	all pass
small test	23,552	47,104	0.9988	0.037	0.022	+0.022	+0.24	+0.19	yes / 0 / 0	yes / 0 / 2	0.69	1.63	self-intersections, ICP noise
sphere	9,024	18,048	0.9997	0.034	0.016	+0.015	+0.06	+0.04	yes / 0 / 0	yes / 0 / 0	1.57	2.84	all pass
test bolus 107mL	15,212	30,424	0.9960	0.110	0.038	+0.007	+0.15	+0.69	yes / 0 / 0	yes / 0 / 0	0.92	2.57	all pass
test bolus 7mm	10,594	21,188	0.9947	0.133	0.039	-0.012	-0.32	+0.29	yes / 0 / 0	yes / 0 / 0	1.15	1.80	all pass

6 of 11 cases pass every default meshcompare check. Every failure is topology, one volume margin, or ICP fitting noise — none is a distance or overlap metric.

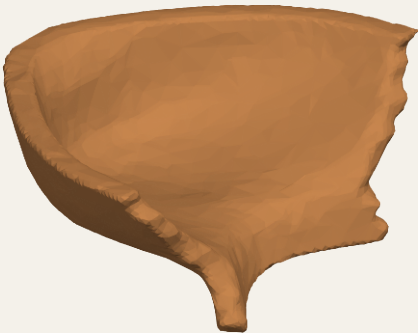
chin_bolus

3,216 input facets · 6,432 facets out of both pipelines · input surface 13,481 mm², volume 30,805 mm³

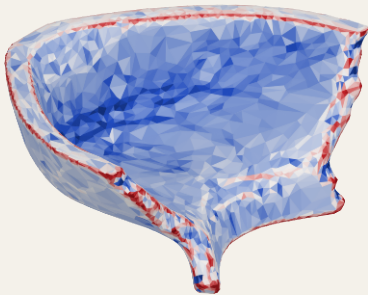
main (MeshLib)



GeometryEngine

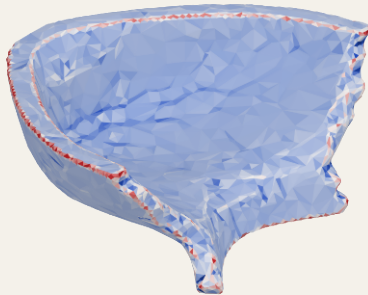


main vs original



mean |dev| 0.118 mm 34% inside

engine vs original



mean |dev| 0.097 mm 19% inside

The smoothed mesh from each pipeline. Every panel on this page shares one camera, fitted to the unsmoothed input.

Metrics

engine vs main

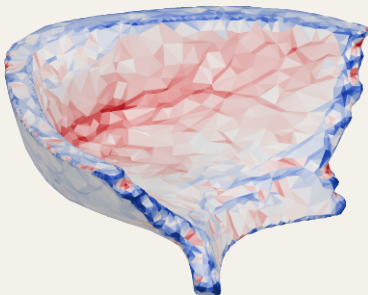
Dice	0.9907
HD95 / Hausdorff	0.133 / 0.801 mm
mean separation (ASSD)	0.0451 mm
signed bias	+0.0071 mm
surface Dice @0.5 mm	0.9999
volume / area	+0.31 % / +1.17 %
curvature ratio	1.233

fidelity to the input

main → engine

Dice	0.9773 → 0.9788
HD95	0.253 → 0.147 mm
mean separation	0.108 → 0.100 mm
signed bias	+0.083 → +0.090 mm
volume	+3.62 → +3.94 %
runtime	1.52 → 2.92 s
ICP residual	0.031 mm, 0.050°

engine vs main



mean |dev| 0.067 mm 33% inside

topology

main

engine

watertight	yes	NO
non-manifold edges	0	3
self-intersections	0	0
components	1	1
genus	0	n/a
degenerate facets	0	0

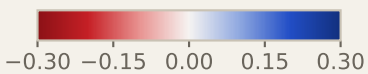
n/a: genus is undefined for a surface this open or this non-manifold.

meshcompare default checks

- `cand.watertight==1` (actual 0)
- `cand.nonmanifold_edges==0` (actual 3)
- `cand.genus==ref.genus` (not computable)

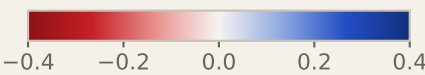
Signed deviation

panel above · ±0.3 mm



Signed deviation

both panels above · ±0.4 mm



Each panel colours the mesh named first in its title by how far it lies from the mesh named second. Red: the coloured surface lies inside the one it is measured against — material the smoothing took away. Blue: outside it — material the smoothing added. Near-white: the two surfaces coincide.

The two fidelity panels share one scale so they can be read against each other. The head-to-head below has its own, because the pipelines differ from each other by far less than either differs from the input.

ear_bolus

2,810 input facets · 5,624 facets out of both pipelines · input surface 11,134 mm², volume 23,914 mm³

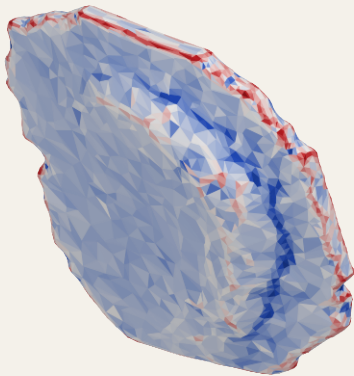
main (MeshLib)



GeometryEngine

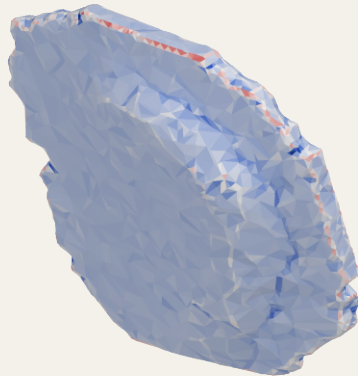


main vs original



mean |dev| 0.127 mm 34% inside

engine vs original



mean |dev| 0.095 mm 15% inside

The smoothed mesh from each pipeline. Every panel on this page shares one camera, fitted to the unsmoothed input.

Metrics

engine vs main

Dice	0.9898
HD95 / Hausdorff	0.183 / 0.697 mm
mean separation (ASSD)	0.0467 mm
signed bias	+0.0121 mm
surface Dice @0.5 mm	0.9999
volume / area	+0.53 % / +1.72 %
curvature ratio	1.284

topology

	main	engine
watertight	yes	yes
non-manifold edges	0	0
self-intersections	0	0
components	1	1
genus	0	0
degenerate facets	0	0

fidelity to the input

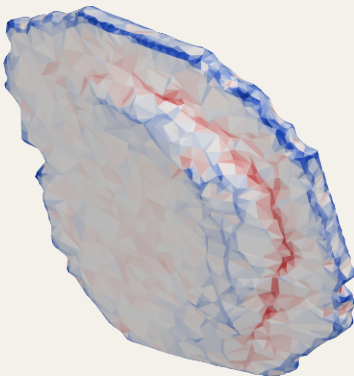
Dice	0.9747 → 0.9776
HD95	0.273 → 0.145 mm
mean separation	0.113 → 0.099 mm
signed bias	+0.082 → +0.094 mm
volume	+3.80 → +4.35 %
runtime	1.02 → 2.09 s
ICP residual	0.008 mm, 0.014°

main → engine

meshcompare default checks

all ten pass

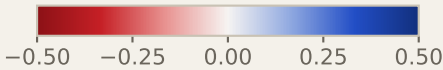
engine vs main



mean |dev| 0.084 mm 35% inside

Signed deviation

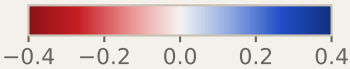
both panels above · ±0.5 mm



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Signed deviation

panel above · ±0.4 mm



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eye_bolus

1,154 input facets · 2,308 facets out of both pipelines · input surface 4,760 mm², volume 9,847 mm³

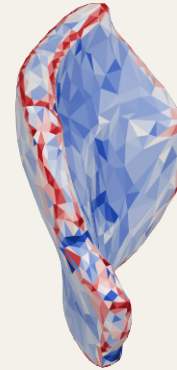
main (MeshLib)



GeometryEngine

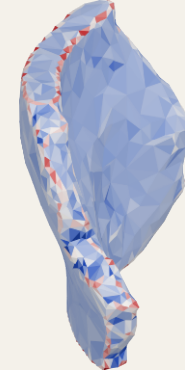


main vs original



mean |dev| 0.122 mm 43% inside

engine vs original



mean |dev| 0.089 mm 21% inside

The smoothed mesh from each pipeline. Every panel on this page shares one camera, fitted to the unsmoothed input.

Metrics

engine vs main

Dice	0.9861
HD95 / Hausdorff	0.226 / 0.814 mm
mean separation (ASSD)	0.0612 mm
signed bias	+0.0295 mm
surface Dice @0.5 mm	0.9997
volume / area	+1.34 % / +2.64 %
curvature ratio	1.270

topology

	main	engine
watertight	yes	yes
non-manifold edges	0	0
self-intersections	0	0
components	1	1
genus	0	0
degenerate facets	0	0

fidelity to the input

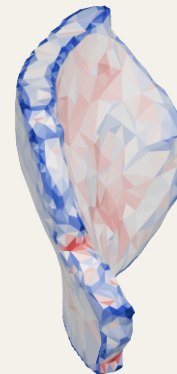
Dice	0.9738 → 0.9773
HD95	0.284 → 0.146 mm
mean separation	0.112 → 0.096 mm
signed bias	+0.059 → +0.087 mm
volume	+2.84 → +4.21 %
runtime	0.63 → 0.85 s
ICP residual	0.018 mm, 0.102°

main → engine

meshcompare default checks

- cand_ref.abs_volume_diff_pct ≤ 1 (actual 1.33772)
- icp.rotation_deg ≤ 0.1 (actual 0.101575)

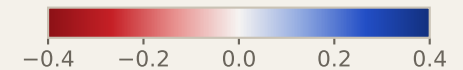
engine vs main



mean |dev| 0.102 mm 24% inside

Signed deviation

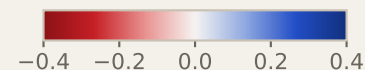
both panels above · ±0.4 mm



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Signed deviation

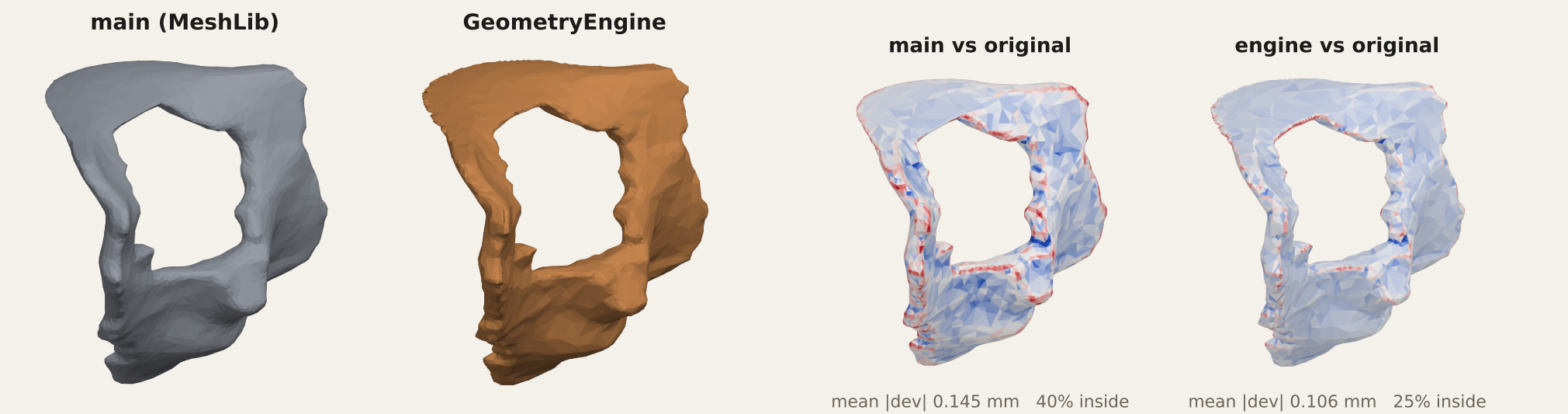
panel above · ±0.4 mm



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larynx small

2,978 input facets · 5,960 facets out of both pipelines · input surface 14,058 mm², volume 28,494 mm³



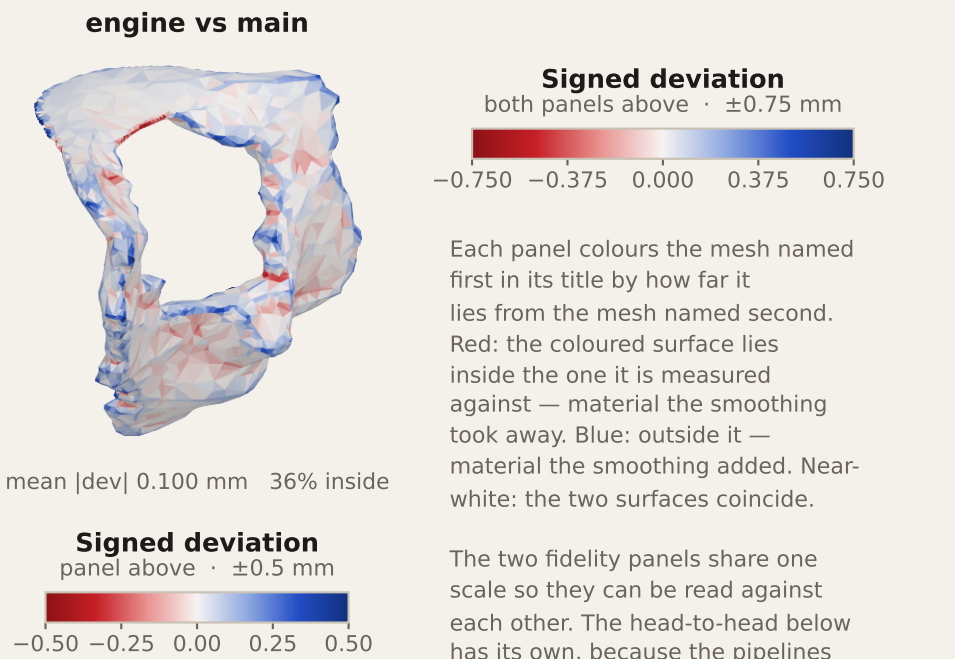
The smoothed mesh from each pipeline. Every panel on this page shares one camera, fitted to the unsmoothed input.

Metrics

engine vs main		fidelity to the input		main → engine	
Dice	0.9844	Dice	0.9705	→	0.9757
HD95 / Hausdorff	0.218 / 1.320 mm	HD95	0.398	→	0.220 mm
mean separation (ASSD)	0.0690 mm	mean separation	0.130	→	0.105 mm
signed bias	+0.0161 mm	signed bias	+0.064	→	+0.080 mm
surface Dice @0.5 mm	0.9983	volume	+3.20	→	+3.97 %
volume / area	+0.74 % / +2.44 %	runtime	0.68	→	2.29 s
curvature ratio	1.228	ICP residual	0.007 mm,	0.022°	

topology	main	engine	meshcompare default checks
watertight	yes	NO	• cand.watertight==1 (actual 0)
non-manifold edges	0	3	• cand.nonmanifold_edges==0 (actual 3)
self-intersections	0	0	• cand.genus==ref.genus (not computable)
components	1	1	
genus	1	n/a	
degenerate facets	0	0	

n/a: genus is undefined for a surface this open or this non-manifold.



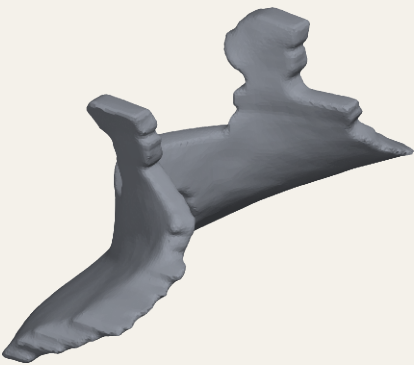
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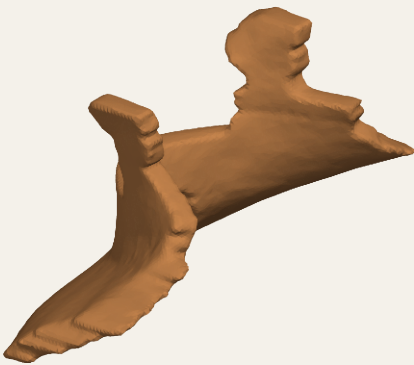
larynx_bolus

10,694 input facets · 21,388 facets out of both pipelines · input surface 45,689 mm², volume 178,457 mm³

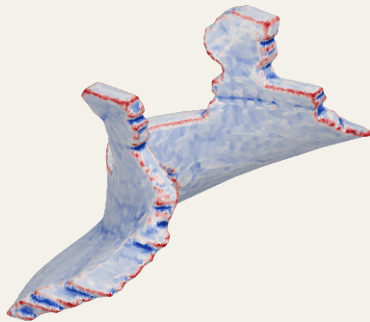
main (MeshLib)



GeometryEngine

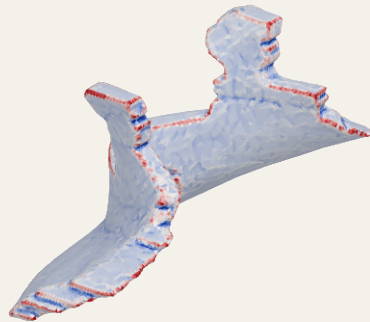


main vs original



mean |dev| 0.141 mm 31% inside

engine vs original



mean |dev| 0.119 mm 23% inside

The smoothed mesh from each pipeline. Every panel on this page shares one camera, fitted to the unsmoothed input.

Metrics

engine vs main

Dice	0.9961
HD95 / Hausdorff	0.121 / 1.228 mm
mean separation (ASSD)	0.0322 mm
signed bias	+0.0015 mm
surface Dice @0.5 mm	0.9990
volume / area	+0.04 % / +0.63 %
curvature ratio	1.029

topology

	main	engine
watertight	yes	NO
non-manifold edges	0	1
self-intersections	0	0
components	1	1
genus	1	n/a
degenerate facets	0	0

n/a: genus is undefined for a surface this open or this non-manifold.

fidelity to the input

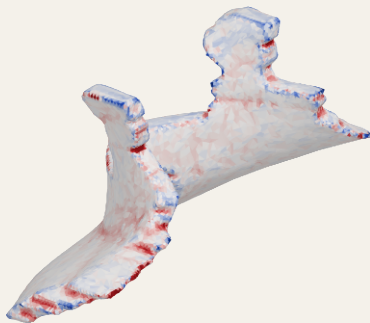
Dice	0.9850 → 0.9863
HD95	0.364 → 0.270 mm
mean separation	0.123 → 0.112 mm
signed bias	+0.070 → +0.072 mm
volume	+1.82 → +1.85 %
runtime	1.43 → 2.50 s
ICP residual	0.004 mm, 0.002°

main → engine

meshcompare default checks

- `cand.watertight==1` (actual 0)
- `cand.nonmanifold_edges==0` (actual 1)
- `cand.genus==ref.genus` (not computable)

engine vs main



mean |dev| 0.049 mm 47% inside

Signed deviation

panel above · ±0.4 mm



Signed deviation

both panels above · ±0.75 mm



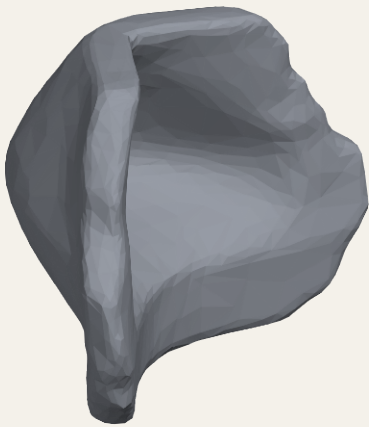
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nose_bolus

1,566 input facets · 3,132 facets out of both pipelines · input surface 6,030 mm², volume 13,294 mm³

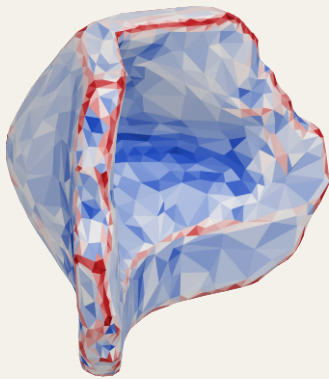
main (MeshLib)



GeometryEngine

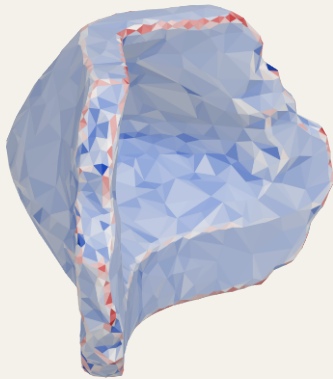


main vs original



mean |dev| 0.134 mm 37% inside

engine vs original



mean |dev| 0.096 mm 17% inside

The smoothed mesh from each pipeline. Every panel on this page shares one camera, fitted to the unsmoothed input.

Metrics

engine vs main

Dice	0.9857
HD95 / Hausdorff	0.200 / 0.639 mm
mean separation (ASSD)	0.0669 mm
signed bias	+0.0119 mm
surface Dice @0.5 mm	0.9999
volume / area	+0.51 % / +2.34 %
curvature ratio	1.228

topology

	main	engine
watertight	yes	yes
non-manifold edges	0	0
self-intersections	0	0
components	1	1
genus	0	0
degenerate facets	0	0

fidelity to the input

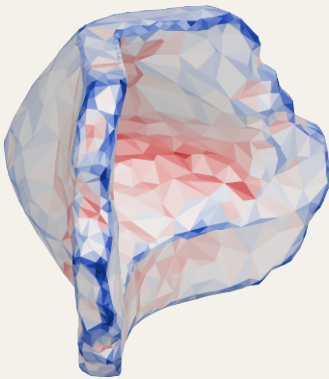
Dice	0.9739 → 0.9783
HD95	0.336 → 0.174 mm
mean separation	0.121 → 0.099 mm
signed bias	+0.080 → +0.091 mm
volume	+3.59 → +4.12 %
runtime	0.67 → 0.98 s
ICP residual	0.030 mm, 0.092°

main → engine

meshcompare default checks

all ten pass

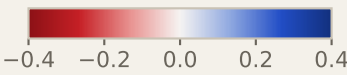
engine vs main



mean |dev| 0.091 mm 35% inside

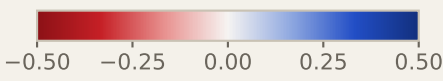
Signed deviation

panel above · ±0.4 mm



Signed deviation

both panels above · ±0.5 mm

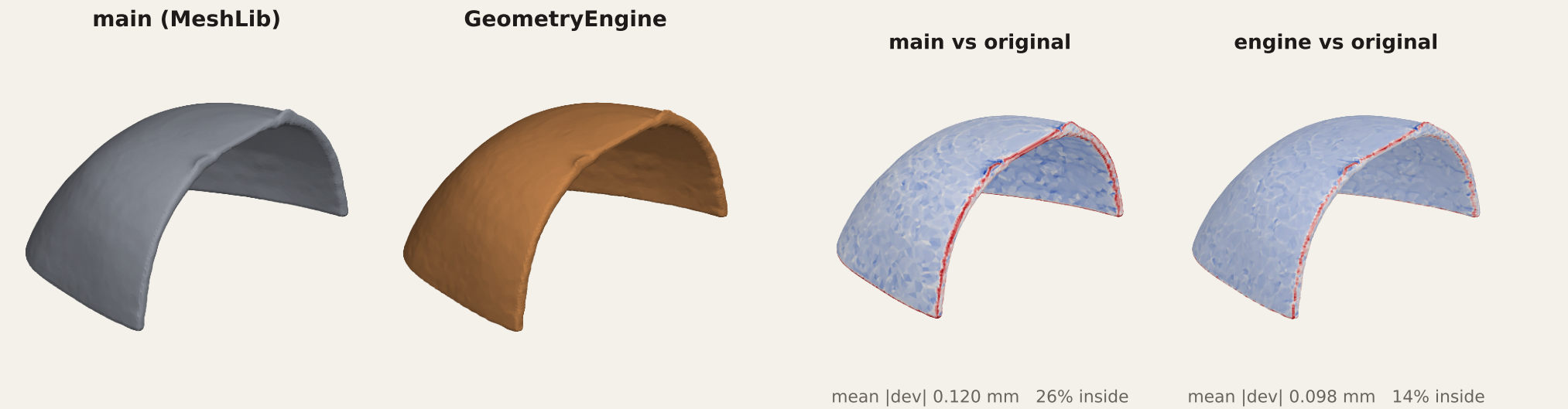


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scalp_bolus

8,546 input facets · 17,092 facets out of both pipelines · input surface 42,347 mm², volume 100,832 mm³



The smoothed mesh from each pipeline. Every panel on this page shares one camera, fitted to the unsmoothed input.

Metrics

engine vs main

Dice	0.9946
HD95 / Hausdorff	0.086 / 0.658 mm
mean separation (ASSD)	0.0270 mm
signed bias	+0.0040 mm
surface Dice @0.5 mm	1.0000
volume / area	+0.16 % / +0.59 %
curvature ratio	1.058

fidelity to the input

main → engine

Dice	0.9778 → 0.9796
HD95	0.193 → 0.138 mm
mean separation	0.109 → 0.100 mm
signed bias	+0.085 → +0.089 mm
volume	+3.58 → +3.74 %
runtime	1.29 → 3.35 s
ICP residual	0.037 mm, 0.023°

topology

main

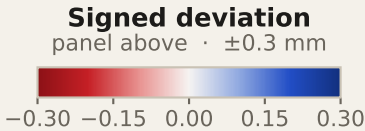
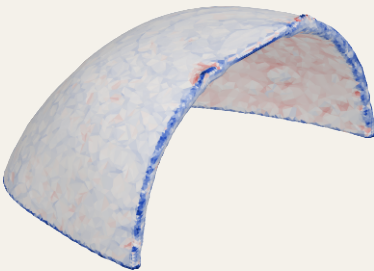
engine

watertight	yes	yes
non-manifold edges	0	0
self-intersections	0	0
components	1	1
genus	0	0
degenerate facets	0	0

meshcompare default checks

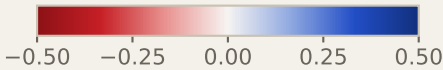
all ten pass

engine vs main



Signed deviation

both panels above · ±0.5 mm



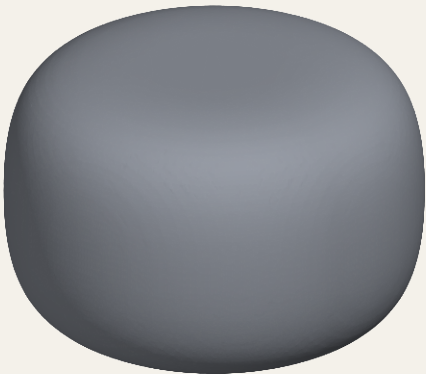
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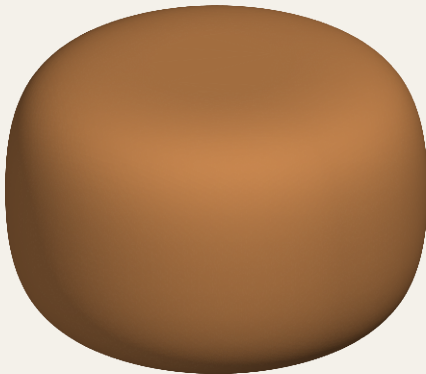
small test

23,552 input facets · 47,104 facets out of both pipelines · input surface 10,575 mm², volume 96,627 mm³

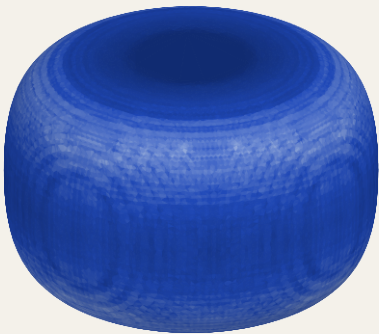
main (MeshLib)



GeometryEngine

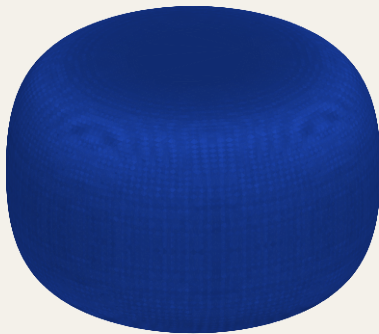


main vs original



mean |dev| 0.062 mm 0% inside

engine vs original



mean |dev| 0.089 mm 0% inside

The smoothed mesh from each pipeline. Every panel on this page shares one camera, fitted to the unsmoothed input.

Metrics

engine vs main

Dice	0.9988
HD95 / Hausdorff	0.037 / 0.056 mm
mean separation (ASSD)	0.0219 mm
signed bias	+0.0219 mm
surface Dice @0.5 mm	1.0000
volume / area	+0.24 % / +0.19 %
curvature ratio	0.810

topology

	main	engine
watertight	yes	yes
non-manifold edges	0	0
self-intersections	0	2
components	1	1
genus	0	0
degenerate facets	0	0

fidelity to the input

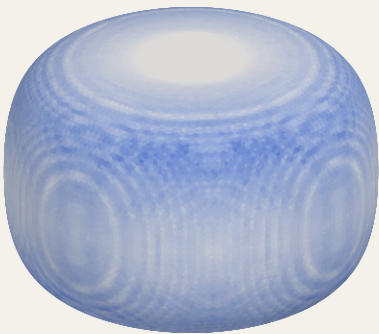
Dice	0.9963 → 0.9951
HD95	0.099 → 0.100 mm
mean separation	0.069 → 0.090 mm
signed bias	+0.069 → +0.090 mm
volume	+0.75 → +0.99 %
runtime	0.69 → 1.63 s
ICP residual	0.001 mm, 0.177°

main → engine

meshcompare default checks

- `cand.self_intersections==0` (actual 2)
- `icp.rotation_deg<=0.1` (actual 0.177261)

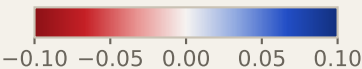
engine vs main



mean |dev| 0.023 mm 0% inside

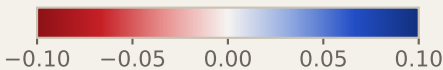
Signed deviation

panel above · ±0.1 mm



Signed deviation

both panels above · ±0.1 mm

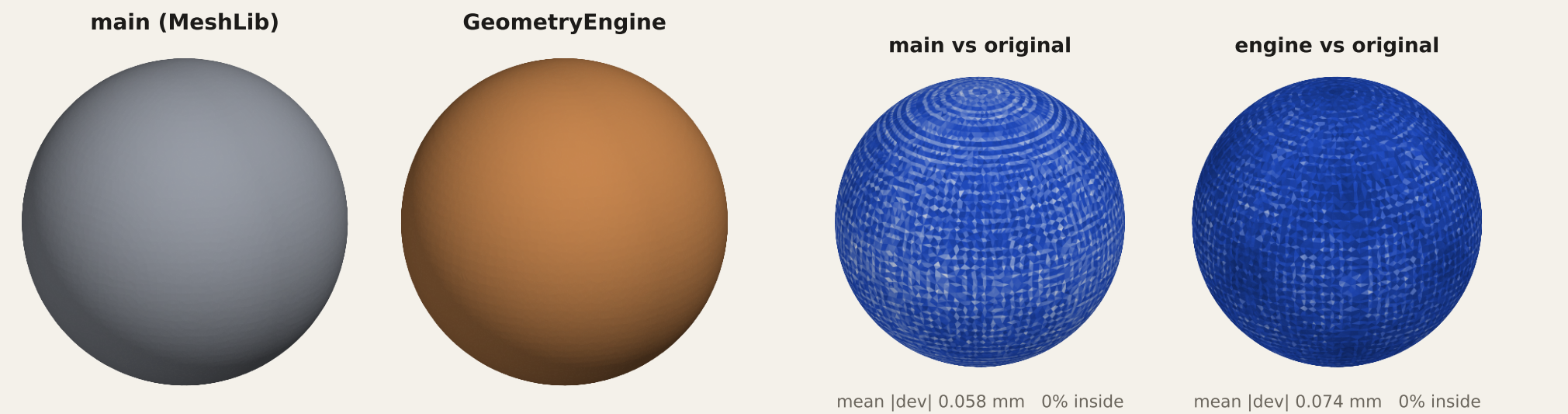


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sphere

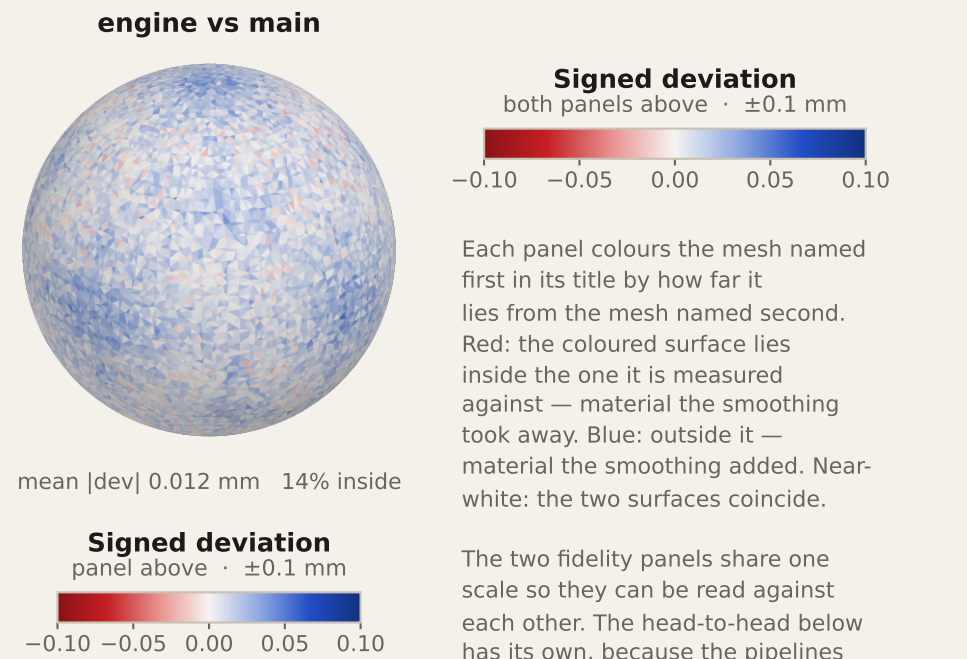
9,024 input facets · 18,048 facets out of both pipelines · input surface 80,353 mm², volume 2,140,815 mm³



The smoothed mesh from each pipeline. Every panel on this page shares one camera, fitted to the unsmoothed input.

Metrics

engine vs main			fidelity to the input		main → engine	
Dice		0.9997	Dice		0.9988	→ 0.9985
HD95 / Hausdorff	0.034 / 0.081 mm		HD95		0.084	→ 0.096 mm
mean separation (ASSD)	0.0161 mm		mean separation		0.064	→ 0.079 mm
signed bias	+0.0154 mm		signed bias		+0.064	→ +0.079 mm
surface Dice @0.5 mm	1.0000		volume		+0.24	→ +0.30 %
volume / area	+0.06 % / +0.04 %		runtime		1.57	→ 2.84 s
curvature ratio	0.963		ICP residual		0.001 mm,	0.076°
topology	main	engine	meshcompare default checks			
watertight	yes	yes	all ten pass			
non-manifold edges	0	0				
self-intersections	0	0				
components	1	1				
genus	0	0				
degenerate facets	0	0				

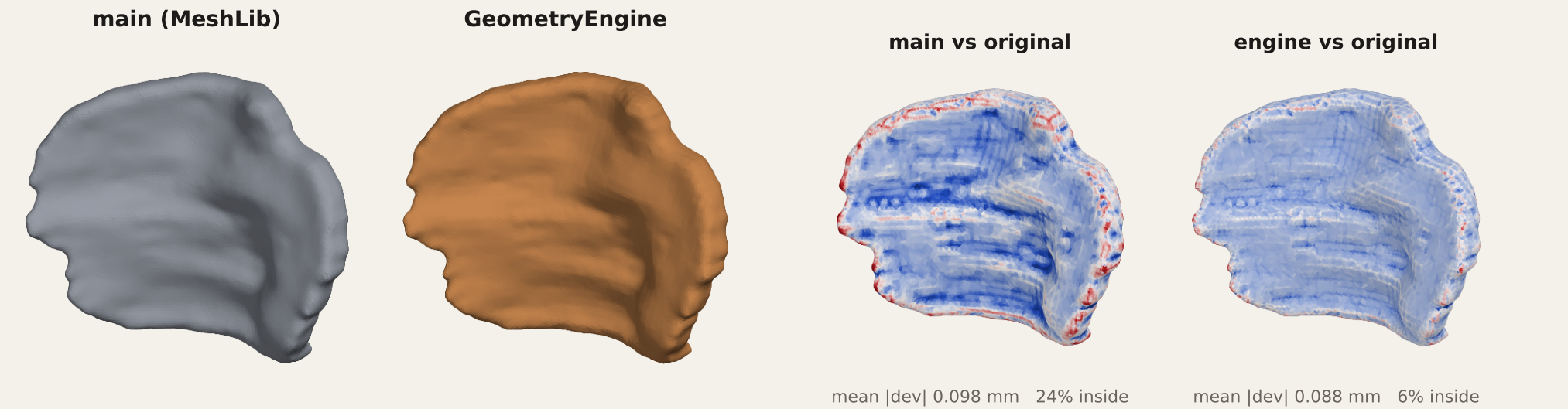


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test bolus 107mL

15,212 input facets · 30,424 facets out of both pipelines · input surface 23,295 mm², volume 106,883 mm³



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Metrics

engine vs main

Dice	0.9960
HD95 / Hausdorff	0.110 / 0.599 mm
mean separation (ASSD)	0.0379 mm
signed bias	+0.0072 mm
surface Dice @0.5 mm	1.0000
volume / area	+0.15 % / +0.69 %
curvature ratio	0.964

fidelity to the input

Dice	0.9893 → 0.9900
HD95	0.224 → 0.144 mm
mean separation	0.100 → 0.093 mm
signed bias	+0.083 → +0.090 mm
volume	+1.80 → +1.96 %
runtime	0.92 → 2.57 s
ICP residual	0.012 mm, 0.008°

main → engine

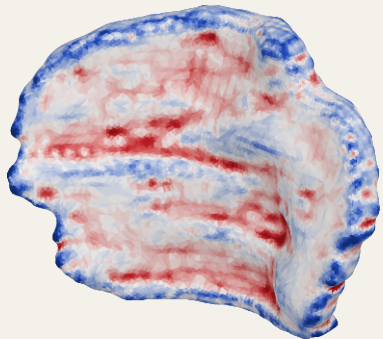
topology

	main	engine
watertight	yes	yes
non-manifold edges	0	0
self-intersections	0	0
components	1	1
genus	0	0
degenerate facets	0	0

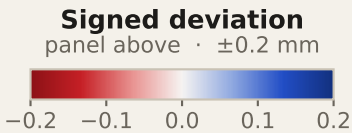
meshcompare default checks

all ten pass

engine vs main



mean |dev| 0.045 mm 37% inside



Signed deviation

both panels above · ±0.4 mm

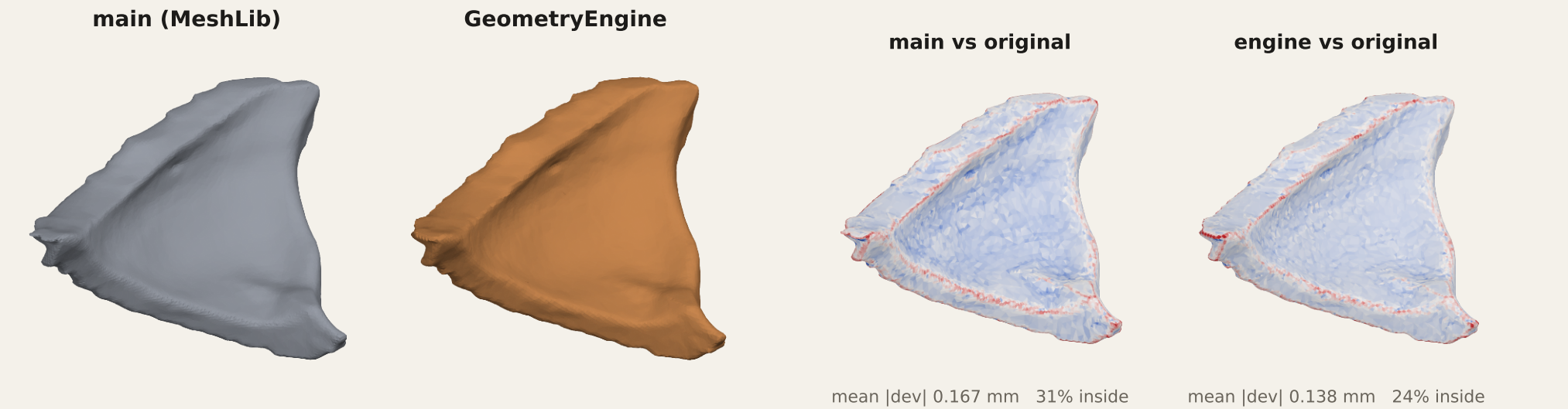


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test bolus 7mm

10,594 input facets · 21,188 facets out of both pipelines · input surface 56,349 mm², volume 191,158 mm³



The smoothed mesh from each pipeline. Every panel on this page shares one camera, fitted to the unsmoothed input.

Metrics

engine vs main

Dice	0.9947
HD95 / Hausdorff	0.133 / 3.425 mm
mean separation (ASSD)	0.0387 mm
signed bias	-0.0121 mm
surface Dice @0.5 mm	0.9987
volume / area	-0.32 % / +0.29 %
curvature ratio	0.920

fidelity to the input

main → engine

Dice	0.9803 → 0.9825
HD95	0.478 → 0.393 mm
mean separation	0.141 → 0.128 mm
signed bias	+0.088 → +0.074 mm
volume	+2.61 → +2.29 %
runtime	1.15 → 1.80 s
ICP residual	0.013 mm, 0.005°

topology

main

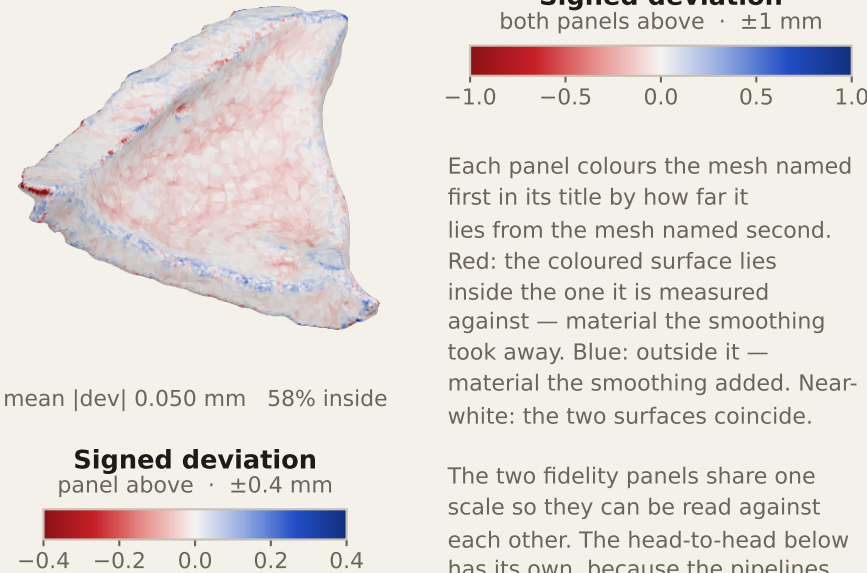
engine

watertight	yes	yes
non-manifold edges	0	0
self-intersections	0	0
components	1	1
genus	0	0
degenerate facets	0	0

meshcompare default checks

all ten pass

engine vs main



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